

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
ELECTRONICS LAB
EXPERIMENT NO.8

Aim : To Study the operation of flip-flops and verify their truth table.

1. "RS" & "D Type Flip Flops using NAND Gates.
2. "D Type Flip Flop using TTL IC.
3. "JK" Flip Flop using TTL IC.
4. "T" Flip Flop

Instruments Required :

- Fixed Output DC Regulated Power supply of 5V.
- A 1Hz Monoshot Clock Pulse with Pulser switch is provided on the front panel.
- Four Logic Inputs Logic '0' & Logic '1' selectable using SPDT switches are provided on the front panel
- Two red output indicators are also provided on the front panel.
- IC 7400, 7410, 7474 & 7476 are mounted on the front panel & important
- Connections are brought out on sockets.

Theory :

A flip flop is an electronic circuit that has two stable states, one representing a binary '1' and the other binary '0'. If put into one state, the flip flop will remain in that state as long as power is applied or until it is changed. It thus remembers the data. In digital circuits, flip flops are used in a variety of storage, counting, sequencing and timing applications. There are three basic types of flip flops the set-reset (also known as R-S flip flop or a latch), the 'D' type and the 'JK'. The 'RS' flip flop is the simplest. It has two inputs 'S' & 'R' and two outputs 'Q' and 'Q'. Applying appropriate logic signals to either 'S' or 'R' input will put the latch into one state or the other. When a flip flop is set by 'S' input, it is said to be storing binary '1' ('Q' output is = High). When reset by 'R' input, it is said to be storing binary '0' ('Q' output = Low).

Like any other flip flop, the 'D' flip flop has two outputs that determine whether it is storing a binary '1' or a binary '0'. It also has two inputs. These are called 'D' and 'T' and work differently. The data or bit to be stored (which can be either a binary '0' or '1') is applied to the 'D' input. The 'T' input line controls the flip flop. It is used to determine whether the input data at 'D' is to be recognized or to be ignored. If the 'T' input is High, the data on the 'D' line gets stored in the flip flop. If the 'T' line is low the 'D' input line data has no effect and the bit stored previously is retained.

The 'JK' flip flop is the most versatile binary storage element. It can perform all the functions of 'R', 'S' and 'D' flip flops plus it can do several other things. An integrated circuit 'JK' flip flop is really two 'RS' flip flops in one. These are called Master and Slave. Both flip flops are controlled by a common clock pulse to the 'T' input. When the 'T' line

goes High, cutting off the Slave. At the same time data on 'J' and 'K' inputs is passed on to the Master for storage. When the 'T' line goes Low, cutting off the Master from the input circuits. At the same time gates 'C' and 'D' are enabled and data stored in the Master is transferred to the Slave. This technique provides a complete isolation between the inputs and outputs. The integrated circuit IC 7476 contains two identical JK flip-flops which are completely independent except for a common power supply input connection.

Procedure :

Verification of 'RS' Flip Flop:

1. Connect the 4 logic inputs to 'Preset (PR)', 'Clear (CLR)', 'S' & 'R' input of the Flip-Flop as shown in Fig. No. (1) through patch cords. Also connect 'Q' & $\bar{Q}$ outputs to output indicators.
2. Connect 1Hz clock output to 'Clock (CK)' input of the flip flop.
3. Switch ON the instrument using ON/OFF toggle switch provided on the front panel.
4. Verify the Truth Table No. (1) for various sets of input combinations.

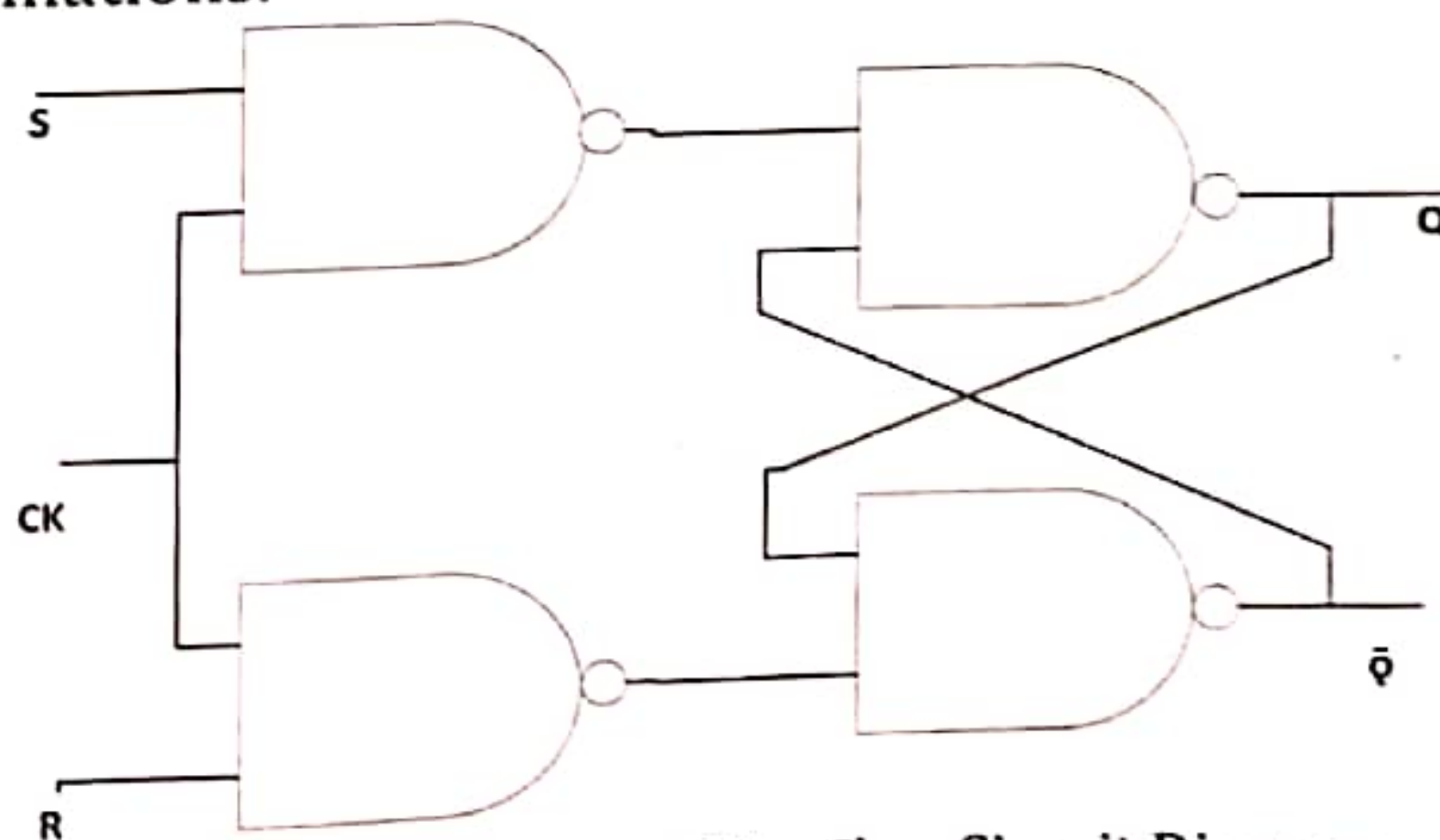


Figure No. (1) 'RS' Type Flip Flop Circuit Diagram

TRUTH TABLE No.1 'RS' FLIP FLOP

INPUTS					OUTPUTS	
PRESET T (PR)	CLEAR (CLR)	CLOCK (CK)	S	R	Q	$\bar{Q}$
L	H	X	L	L	H	L
H	L	X	L	L	L	H
L	L	X	L	L	H	H
H	H	P	L	L	Q	Q
H	H	P	H	L	H	L
H	H	P	L	H	L	H
H	H	P	H	H	TOGGLE	TOGGLE

Verification of 'D' Type Flip Flop:

1. Connect the output of NOT Gate to "R" input through patch cord as shown in Fig. No. (2). Connect 3 logic inputs to 'Preset (PR), Clear (CLR)' & 'D' input of the Flip-Flop as shown in Fig. No. (2) through patch cords. Also connect 'Q' & 'Q' outputs to output indicators.
2. Connect 1Hz clock output to 'Clock (CK)' input of the flip flop.
3. Switch ON the instrument using ON/ OFF toggle switch provided on the front panel.
4. Verify the Truth Table No. (2) for various sets of input combinations.

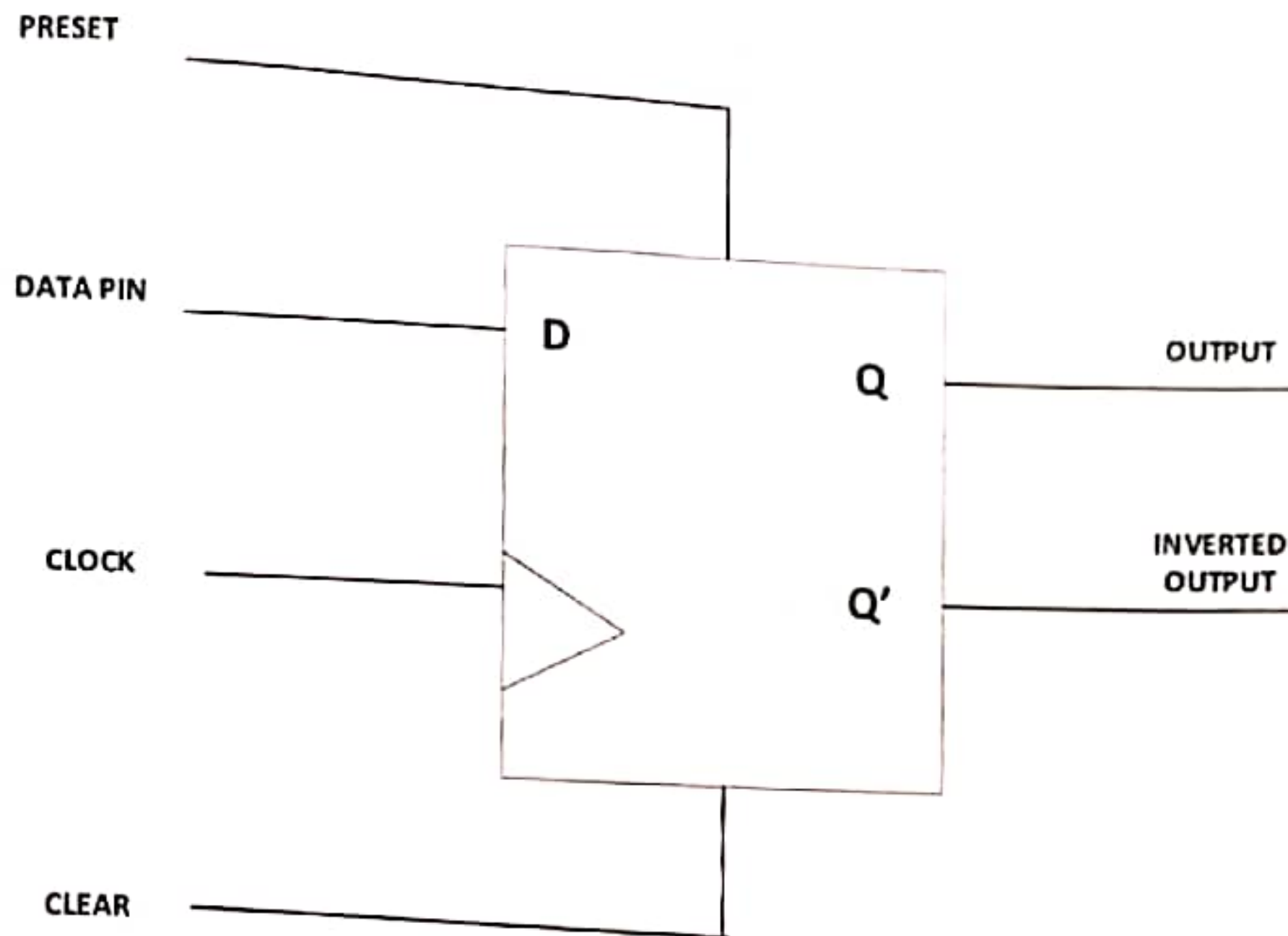


Figure No. (2) 'D' Type Flip Flop Circuit Diagram

TRUTH TABLE No. (2) 'D' FLIP FLOP

INPUTS				OUTPUTS	
PRESET (PR)	CLEAR (CLR)	CLOCK (CK)	D	Q	$\bar{Q}$
L	H	X	L	H	L
H	L	X	L	L	H
L	L	X	L	H	H
H	H	P	H	H	L
H	H	P	L	L	H

Verification of 'JK' Flip Flop:

1. Connect the 4 logic inputs to 'Preset (PR)', Clear (CLR)', 'J' & 'K' input of the Flip-Flop as shown in Fig. No. (3) through patch cords. Also connect 'Q' & 'Q' outputs to output indicators.
2. Connect 1Hz clock output to 'Clock (CK)' input of the flip flop.
3. Switch ON the instrument using ON/OFF toggle switch provided on the front panel.
4. Verify the Truth Table No. (3) for various sets of input combinations.

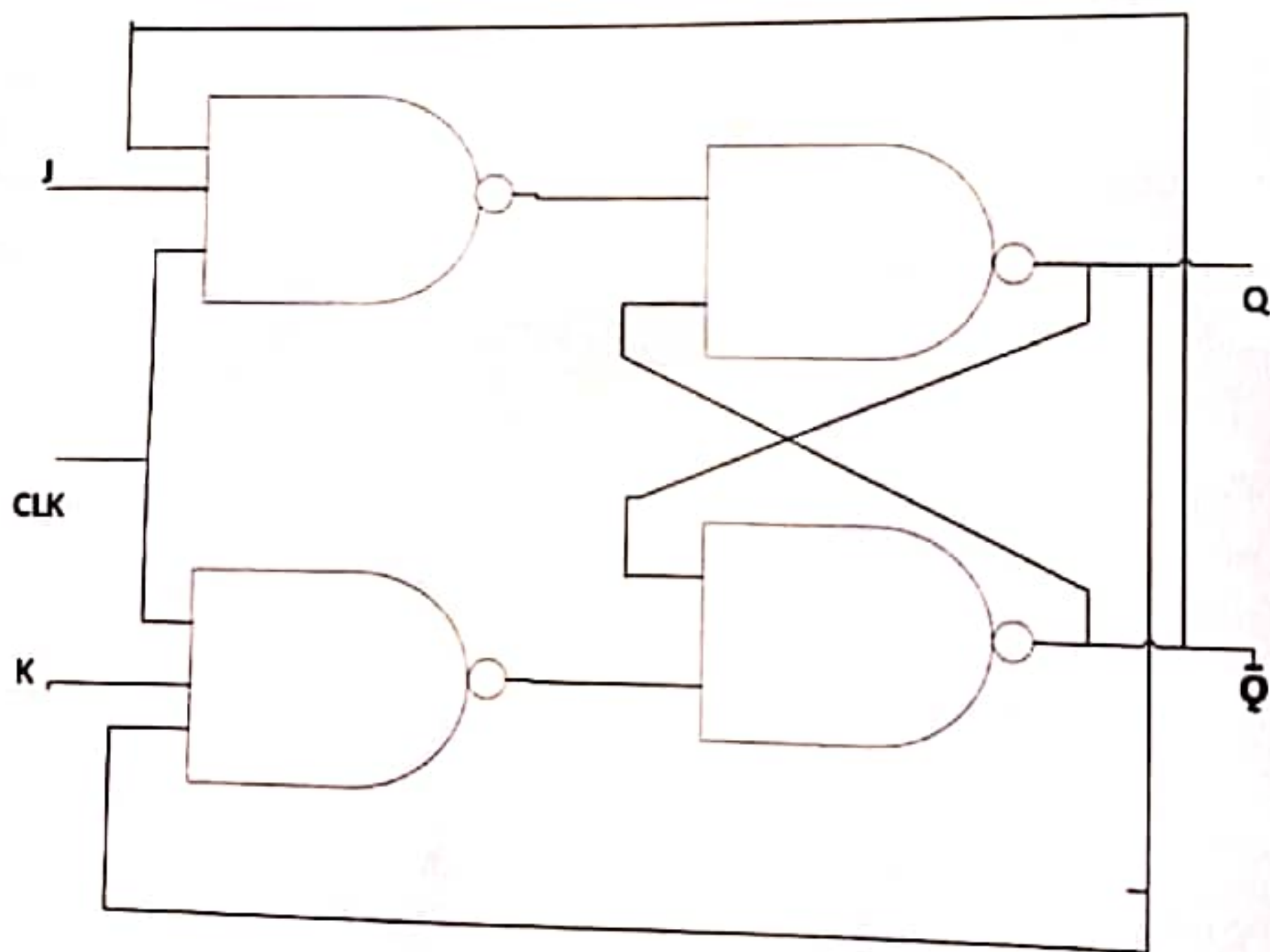


Figure No. (3) 'JK' Type Flip Flop Circuit Diagram

TRUTH TABLE No. (3) 'JK' FLIP FLOP

INPUTS					OUTPUTS	
PRESET (PR)	CLEAR (CLR)	CLOCK (CK)	J	K	Q	$\bar{Q}$
L	H	X	L	L	H	L
H	L	X	L	L	L	H
L	L	X	L	L	H	H
H	H	P	L	L	Q_0	$\bar{Q}_0$
H	H	P	H	L	H	L
H	H	P	L	H	L	H
H	H	P	H	H	TOGGLE	TOGGLE

Verification of 'T' Type Flip Flop:

1. Short the "J" & "K" input of the IC 7476 to form "T" input. Also connect three logic inputs to 'Preset (PR)', Clear (CLR)' & 'T' inputs of the Flip-Flop (To obtain "T" input short the 'J' & 'K' inputs). Also connect 'Q' & 'Q' outputs to output indicators.
2. Connect the logic high input to Clear & Reset.
3. Connect 1Hz clock output to 'Clock (CK)' input of flip flop.
4. Switch ON the instrument using ON/OFF toggle switch provided on the front panel.
5. Verify the Truth Table No. (4) for various sets of input combinations.

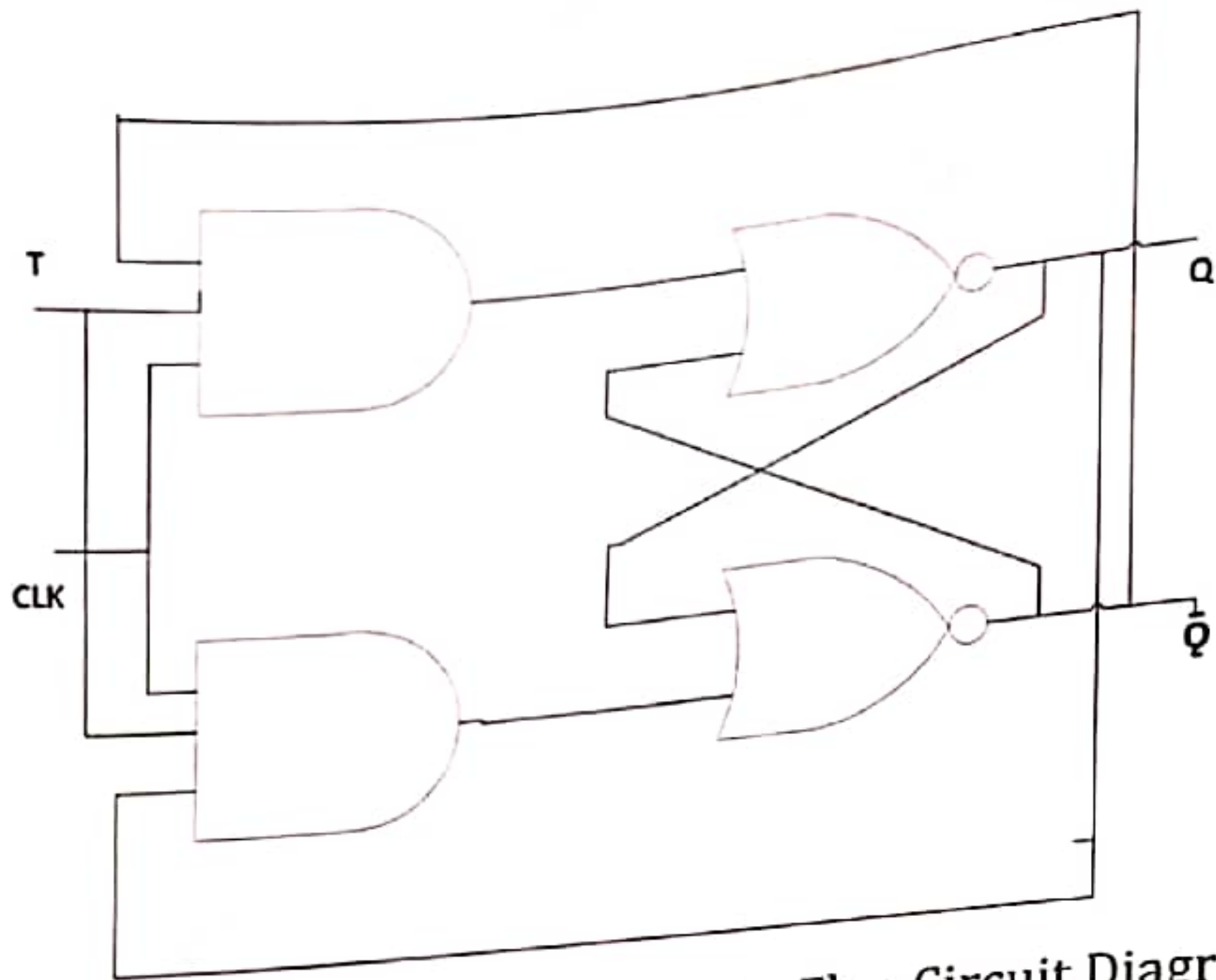


Figure No. (4) 'T' Type Flip Flop Circuit Diagram

TRUTH TABLE No. (4) 'T' TYPE FLIP FLOP

INPUT T	OUTPUT	
	Q	$\bar{Q}$
1	TOGGLE	

Verification of 'D' Type Flip Flop:

1. Connect 3 logic inputs to 'Preset (PR)', Clear (CLR)' & 'D' input of the Flip-Flop as shown in Fig. No. (4) through patch cords. Also connect 'Q' & 'Q' outputs to output indicators.
2. Connect 1Hz clock output to 'Clock (CK)' input of the flip flop.

3. Switch ON the instrument using ON/OFF toggle switch provided on the front panel.
4. Verify the Truth Table No. (5) for various sets of input combinations.

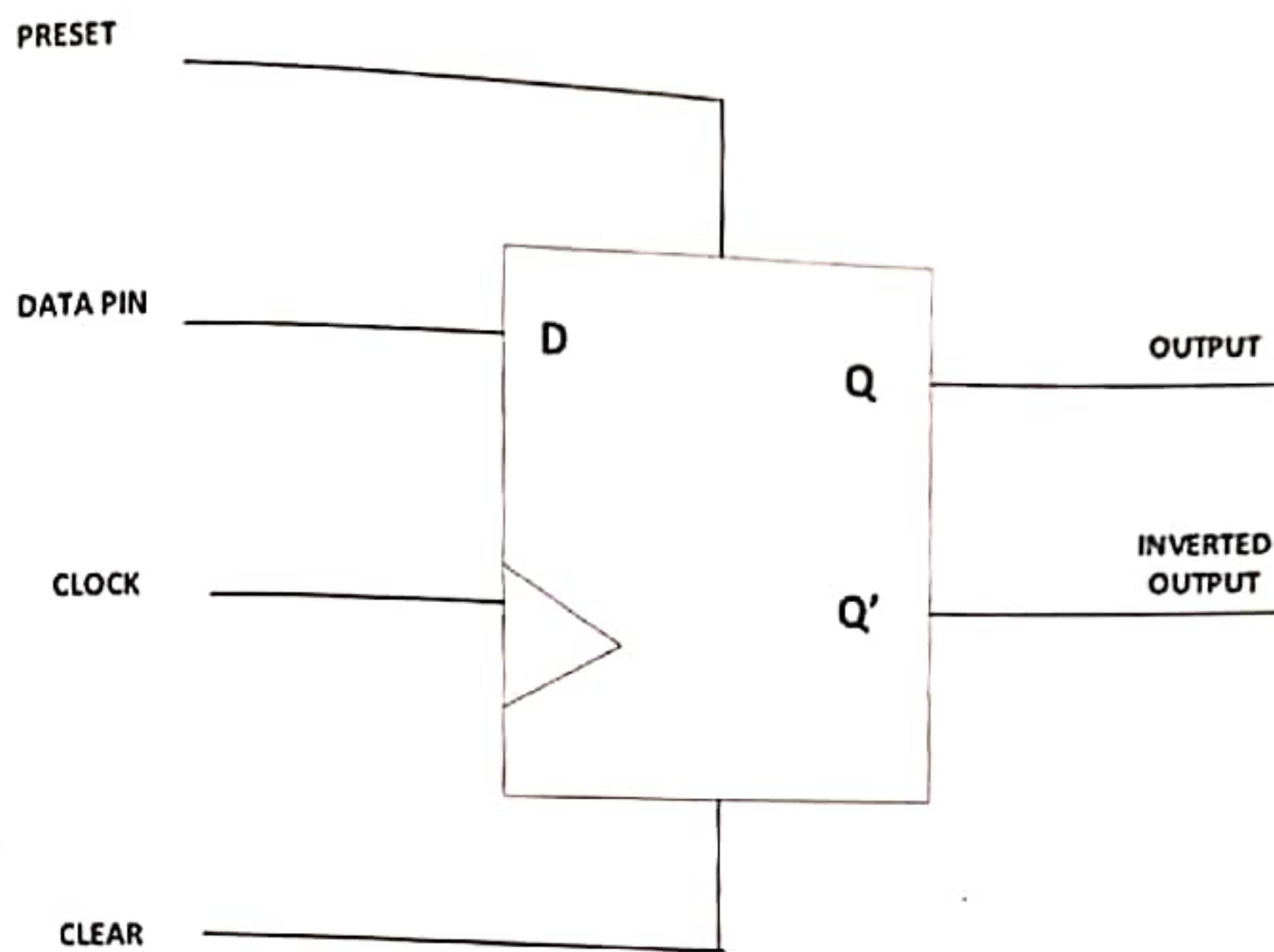


Figure No. (5) 'D' Type Flip Flop Circuit Diagram

TRUTH TABLE No. (5) 'D' FLIP FLOP

INPUTS				OUTPUTS	
PRESET (PR)	CLEAR (CLR)	CLOCK (CK)	D	Q	$\bar{Q}$
L	H	X	L	H	L
H	L	X	L	L	H
L	L	X	L	H	H
H	H	P	H	H	L
H	H	P	L	L	H

STANDARD ACCESSORIES

- | | |
|--|--------|
| 1. Single point (4mm) Patch cords for Interconnection. | 7 Nos. |
| 2. Multipoint (4mm) Patch cords for Interconnection. | 1 No. |
| 3. Instruction Manual | 1 No. |

Note:

- "L" stands for Low input/output.
- "H" stands for High input/output.
- "P" stands for at the application of Clock Pulse.
- "X" doesn't press the clock pulser switch (low value).

Result & Conclusion:

Questions

1. How do flip-flops store data and what is their significance in digital circuits?
2. What are the advantages of using NAND gates to construct flip-flops?
3. How is an RS flip-flop constructed using NAND gates?
4. How is a D flip-flop constructed using NAND gates?
5. What are the advantages of using TTL ICs for constructing D flip-flops?
6. What is the advantage of using a JK Flip-Flop over an RS Flip-Flop?
7. What are the applications of a T Flip-Flop in digital circuits?